

Jaiaditya Asudani

Github: github.com/Jai2379
Portfolio: [Jai2379.github.io](https://jai2379.github.io)

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EDUCATION

- University of Waterloo** Waterloo, Canada
 - Bachelor of Applied Science - Nanotechnology Engineering* *Expected 2030*
 - Courses: Linear Algebra & Calculus for Engineers, Materials Science & Nanomaterials, Electricity, Magnetism & Circuits*

SKILLS SUMMARY

- Languages:** Python, MATLAB
- Computational Tools:** NumPy, SciPy, Streamlit
- Technical Skills:** AutoCAD, OnShape, Solidworks, CATIA, Fusion360, MS Office
- Soft Skills:** Fast Learner, Committed, Highly Motivated, Problem-Solving, Time Management, Attention-to-Detail

EXPERIENCE

- Waterloop - Design Team** *October 2025 - Present*
 - Mechanical Engineering Member*
 - Full-Scale Hyperpod Assembly:** Contributed to system-level CAD integration of the full hyperpod assembly using Onshape and SolidWorks, ensuring proper fit, alignment, and interface compatibility.
 - Mechanical Design & CAD:** Designed and iterated mechanical components in SolidWorks and Onshape, supporting subsystem development and design optimization
 - Materials Research (Carbon Fiber):** Conducted research on carbon fiber and composite materials, contributing to lightweight structural design considerations.
- Electrium Mobility - Design Team** *October 2025 - Present*
 - Mechanical Engineering Member*
 - System-Level Mechanical Construction:** Contributed to the F25 electric bike through CAD modeling and assembly in SolidWorks, supporting structural integration and subsystem fit.
 - Human-Machine Interface Components:** Designed and integrated throttle control housing, handlebar components, and display mounts in SolidWorks, prioritizing rider comfort & interaction.
 - Design for Manufacturability:** Considered manufacturability and assembly constraints during mechanical construction to support reliable prototyping and system robustness
- Mathematics Tutor** *February 2025 - May 2025*
 - Private/Independent*
 - Problem-Solving Mentorship:** Helped student develop logical reasoning and structured approaches to solving high-school level mathematics problems
 - Individualized Instruction:** Provided consistent one-on-one mathematics tutoring, tailoring explanations to address specific learning gaps with a 13% increase in the student's mathematics grade.
 - Professional Responsibility:** Maintained lesson planning, scheduling, and progress tracking over a multi-month period

PROJECTS

- TENG Device (Tribo-Electric Nano Generator):** Designed and fabricated a triboelectric nanogenerator (TENG) during a rapid, hands-on engineering challenge emphasizing real-time prototyping and design improvisation. Demonstrated functional mechanical-to-electrical energy conversion as a proof-of-concept under tight time and resource constraints. Generated 22 V of electrical output.
- 6-DOF Robotic Arm (Prototype):** Engineered a 6-degree-of-freedom robotic arm in SolidWorks, developing a complete mechanical assembly with articulated joints and defined motion constraints. Structured the design to support future Arduino-based actuation and control integration, emphasizing modularity and system feasibility.
- Adaptive Presence-Based Lighting Control System (WIP):** Building a smart desk lighting prototype that monitors user activity via infrared motion sensing and dynamically controls lighting to improve energy efficiency during periods of inactivity. - Arduino Uno, PIR motion sensor, digital I/O, embedded C++, breadboard prototyping.
- Interactive Data Analysis Tool - Python & Streamlit:** Developed an interactive histogram and implemented dynamic binning, distribution fitting, and user-controlled parameters to support exploratory data analysis and statistical insight.

VOLUNTEER EXPERIENCE

- Royal Canadian Air Cadet Squadron** Brampton, Ontario
 - Taught and led a batch of 30+ students on aviation and leadership.* *September 2020 - June 2025*
- IGNITE Stem** Brampton, Ontario
 - Made STEM more accessible for students, held hackathons and other events.* *September 2025 - June 2025*